

HOME TASKS-II

Task 1:(Easy-level)

Given a table of candidates and their skills, you're tasked with finding the candidates best suited for an open Data Science job. You want to find candidates who are proficient in Python, Tableau, and PostgreSQL.

Write a query to list the candidates who possess all of the required skills for the job. Sort the output by candidate ID in ascending order.

Assumption:

- There are no duplicates in the **candidates** table.

candidates Table:

Column Name	Type
candidate_id	integer
skill	varchar

candidates Example Input:

candidate_id	skill
123	Python
123	Tableau
123	PostgreSQL
234	R
234	PowerBI
234	SQL Server
345	Python
345	Tableau

Solution: select candidate_id from candidates
where skill in
('Python','Tableau','PostgreSQL') group by
candidate_id

having count(distinct skill)=3

order by candidate_id;

DataLemur ← All questions < > ⌚ 4:16 || ↺

LinkedIn Interview Question

Question Solution Discussion Submissions

Accepted

Congrats 🎉 - Share this problem, and your solution, on LinkedIn or Twitter!

In your post, don't forget to tag Nick Singh, so that he can comment on and share your post with his audience of 150k+ followers on [LinkedIn](#) and 25k+ followers on [Twitter](#) (which will give your post and profile more visibility)!

Output

candidate_id
123

Expected

candidate_id
123

TIME STATUS YOUR SUBMISSION RUNTIME

07/14/2026 17:49	Solved	Copy To Clipboard	PostgreSQL 14
------------------	--------	-------------------	---------------

Input PostgreSQL 14

```
1 select candidate_id
2 from candidates
3 where skill in ('Python','Tableau','PostgreSQL')
4 group by candidate_id
5 having count(distinct skill)=3
6 order by candidate_id;
7
```

Output

candidate_id
123

Run Code Submit

Task-2: Assume you're given two tables containing data about Facebook Pages and their respective likes (as in "Like a Facebook Page").

Write a query to return the IDs of the Facebook pages that have zero likes. The output should be sorted in ascending order based on the page IDs.

pages Table:

Column Name	Type
page_id	integer
page_name	varchar

pages Example Input:

page_id	page_name
20001	SQL Solutions
20045	Brain Exercises
20701	Tips for Data Analysts

page_likes Table:

Column Name	Type
user_id	integer
page_id	integer
liked_date	datetime

page_likes Example Input:

user_id	page_id	liked_date
111	20001	04/08/2022 00:00:00
121	20045	03/12/2022 00:00:00
156	20001	07/25/2022 00:00:00

Solution: select p.page_id from pages p left
join page_likes pl
on p.page_id=pl.page_id where
pl.page_id is NULL
order by p.page_id;

The screenshot shows the DataLemur interface for a SQL interview question titled "Page With No Likes", sourced from Facebook and marked as "Easy". The question is "Facebook SQL Interview Question". The status is "Accepted". A congratulatory message encourages sharing the problem on LinkedIn or Twitter. The input table is "pages" with columns "page_id" and "liked_date". The expected output is a list of "page_id" values: 20701 and 32728. The user's solution is a SQL query: "select p.page_id from pages p left join page_likes pl on p.page_id=pl.page_id where pl.page_id is NULL order by p.page_id;". The output of the query is also shown as 20701 and 32728.

Page With No Likes
Facebook SQL Interview Question

Accepted

Congrats 🎉 - Share this problem, and your solution, on LinkedIn or Twitter!

In your post, don't forget to tag Nick Singh, so that he can comment on and share your post with his audience of 150k+ followers on [LinkedIn](#) and 25k+ followers on [Twitter](#) (which will give your post and profile more visibility)!

Input

page_id
20701
32728

Expected

page_id
20701
32728

Input PostgreSQL 14

```
1 select p.page_id from pages p
2 left join page_likes pl
3 on p.page_id=pl.page_id
4 where pl.page_id is NULL
5 order by p.page_id;
```

Output

page_id
20701
32728

Run Code Submit

Task-3: Tesla is investigating production bottlenecks and they need your help to extract the relevant data. Write a query to determine which parts have begun the assembly process but are not yet finished.

Assumptions:

- **parts_assembly** table contains all parts currently in production, each at varying stages of the assembly process.
- An unfinished part is one that lacks a **finish_date**.

parts_assembly Table

Column Name	Type
part	string
finish_date	datetime
assembly_step	integer

parts_assembly Example Input

part	finish_date	assembly_step
battery	01/22/2022 00:00:00	1
battery	02/22/2022 00:00:00	2
battery	03/22/2022 00:00:00	3
bumper	01/22/2022 00:00:00	1
bumper	02/22/2022 00:00:00	2
bumper		3
bumper		4

Solution: select part,assembly_step from
parts_assembly
where finish_date is null;

 DataLemur

← All questions

<

>

⌚ 0:00

✓

✕

Unfinished Parts

Tesla SQL Interview Question

Question

Solution

Discussion

Submissions

Accepted

Congrats 🎉 - Share this problem, and your solution, on LinkedIn or Twitter!

In your post, don't forget to tag Nick Singh, so that he can comment on and share your post with his audience of 150k+ followers on [LinkedIn](#) and 25k+ followers on [Twitter](#) (which will give your post and profile more visibility)!

Output

part	assembly_step
bumper	3
bumper	4
engine	5

Expected

part	assembly_step
bumper	3
bumper	4
engine	5

TIME	STATUS	YOUR SUBMISSION	RUNTIME
07/14/2026 18:11	Solved	Copy To Clipboard	PostgreSQL 14

▲ Sourced from

● Tesla

Difficulty

Easy

Input

PostgreSQL 14

```
1 select part,assembly_step
2 from parts_assembly
3 where finish_date is null;
```

Output

part	assembly_step
bumper	3
bumper	4
engine	5

Run Code

Submit